

CUSP: CUSUM-Governed Survival Hazard Alarms at the Perception Onset for Off-Road Navigation

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Abstract—Off-road navigation exposes a robot to potentially hazardous terrain en route. Although learning-based navigation uses safety supervision to choose which path to drive, it provides no runtime alarm when the robot following that path is heading into danger. Such an alarm must be learned from field logs, where human intervention preempts the failure and the failure itself is therefore never observed. The human judges driving unsafe early but typically intervenes only once failure is clearly near, so the intervention marks that judgment late. That earlier judgment is what a runtime alarm must detect, yet no prior intervention-supervised method has targeted it. To address this problem, we introduce CUSP (CUSUM-governed Survival model of the Perception onset), a model-agnostic runtime hazard alarm that learns this moment from intervention-terminated logs. “*Cusp*” is a word for the point at which one state is about to turn into another, and the moment we target is exactly such a cusp: the point at which safe driving turns unsafe in a human’s judgment. We call this point the *perception onset* and annotate it separately from the intervention. A visual hazard head is trained on the annotated onset with a discrete-time survival objective so that driving with and without an onset both supervise the head, and a CUSUM accumulates the predicted onset risk into alarms. We evaluate CUSP at five unseen sites, two autonomous and three teleoperated, with 142 events and every method tuned to the same rate of ten false alarms per hour. CUSP detected 85 events compared to 26 for the best of nine adapted baselines, and the margin comes from hazards to which every signal in the navigation model is blind.

I. INTRODUCTION

The safety issue in learning-based off-road navigation is handled within the navigation policy. Visual methods [1], [2] predict how the terrain will affect the robot from the camera image ahead. Experience-based methods [3]–[5] learn that outcome from the robot’s own driving. Uncertainty-aware models [6]–[8] raise the cost of terrain they have not seen. Risk-aware planners [9] optimize the tail of the cost distribution, and learned policies score sampled trajectories with a critic [10] or push them away from estimated obstacles [11]. In every case, the safety issue is addressed before a path is committed to by choosing among candidates. Once the robot is driving the chosen path, however, no component judges whether continuing it is still safe, and even when every candidate is hazardous, the navigation model still selects the least hazardous one and continues to generate paths without any signal that the situation has changed (Fig. 1). The navigation model therefore leaves a gap at runtime that only an alarm can fill, and none of these methods raises one.

Runtime failure detection has recently been studied for manipulation and vision–language–action policies [12]–[16], but under bounded episodes with observable outcomes and repeatable resets. Continuous off-road driving offers neither. A

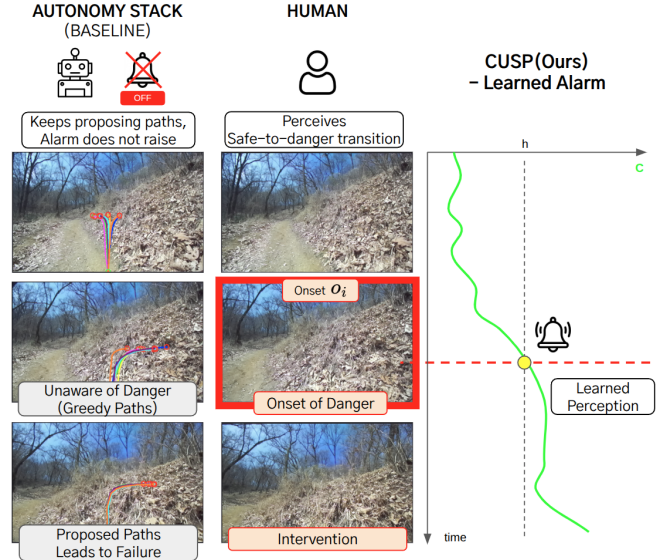


Fig. 1: The challenge this paper addresses. **(Left)** The autonomy model keeps proposing candidate paths as the robot approaches a deteriorating slope, and nothing in the model raises an alarm. **(Middle)** The human perceives the safe-to-dangerous transition, marked as the onset o_i (red box), and takes over before failure. **(Right)** CUSP learns this perception. It accumulates evidence of the transition over time and raises an alarm once that evidence exceeds a threshold, here shortly after the onset.

run has no natural endpoint, and failures cannot be reproduced at will. Simulation has advanced considerably, yet it still does not capture the full range of robot–terrain interaction, and a real rollover or immobilization can end a field campaign. What the logs contain instead is the human’s intervention, which preserves the approach to a hazard while censoring the failure it prevents, so interventions are far more plentiful than failures.

Prior work has used these interventions as supervision. LaND [17] predicts the upcoming intervention from the current observation, and PAAD [18] predicts failure states along a planned path, but both targets are late: the intervention arrives only after the human has recognized the hazard and reacted, and the failure state comes later still.

We instead target the moment at which the human’s judgment shifts from safe to unsafe to continue, and we call this moment the *perception onset*. The intervention does not mark this moment, because by the time the human takes over, the robot has already been in a state severe enough to

require intervention for some time. The human may watch the situation develop before deciding that intervention is necessary, so a label anchored at the intervention describes driving that has been unsafe for a while rather than driving that is turning unsafe. The onset is a human-dependent label rather than an objective safety boundary, but a runtime alarm should reproduce this judgment, and the onset precedes the intervention-based targets discussed above.

Two properties of field logs shape the alarm. Most of the recording captures uneventful driving, and the few onsets it contains are sparse. Discrete-time survival analysis [19], [20] uses both kinds of record. It models the probability that the onset falls in each upcoming interval given that it has not occurred yet. So a segment that reaches an onset supervises when it happened and a segment that ends without one supervises that it had not happened yet. A CUSUM [21] then converts the predicted risk into an alarm. It accumulates the excess over the nominal level frame by frame, so isolated spikes decay and only a sustained rise reaches the threshold.

CUSP (CUSUM-governed Survival model of the Perception onset) implements this in two parts. A lightweight hazard head over frozen vision-foundation features estimates an onset-risk score at every frame. A one-sided CUSUM then accumulates that score into an alarm, with the threshold set from the site's own normal driving to meet a chosen false-alarm rate. Unlike prior intervention-supervised alarms [17], [18], [22], CUSP is supervised by the human's unsafe-to-continue judgment itself and treats the onset as a time-to-event target rather than a binary label. To our knowledge, it is the first runtime alarm to annotate this onset separately from the intervention and learn it as a time-to-event target.

We make three contributions.

- We define the *perception onset*, the moment at which the human judges continued autonomy unsafe, and annotate it separately from the intervention, which no intervention-supervised alarm has done before.
- We propose CUSP, which supervises a visual hazard head on the onset with a discrete-time survival objective and turns the predicted onset risk into alarms with a one-sided CUSUM at a false-alarm rate set from the site's normal driving.
- We show at five unseen sites under a common target false-alarm budget that CUSP detects 85 of 142 events against 26 for the best of nine adapted baselines.

II. RELATED WORK

A. Safety in Off-Road Navigation

The traversability and policy methods surveyed in Sec. I decide which path to take, and none of them judges whether the path already committed to remains safe to continue. Vehicle-centric monitors based on zero-moment-point criteria [23] and terramechanic assurance [24] rest on modeled criteria and each targets one hazard mechanism, and anomaly-based traversability estimation [25] flags terrain that differs from previously traversed terrain, but novelty and hazard do not coincide, since unfamiliar terrain may be safe and familiar terrain

may become unsafe through robot-terrain interaction. None of these approaches uses the human's judgment as supervision.

B. Supervision for Learned Runtime Alarms

Runtime alarms for learned manipulation and vision-language-action policies detect task-contextual anomalies [12] or flag failure from task progress, calibration on successful rollouts, action distributions, or internal representations [13]–[16] – all on finite-horizon rollouts with an explicit end and a success or failure label, which continuous off-road driving lacks.

What off-road driving does have is the human's intervention. LaND [17] predicts the upcoming intervention from the current observation and candidate actions; PAAD [18] defines binary failure states and predicts future failure states from perception and a planned path. GAFT [22] adapts a vision foundation model on frames from intervention-verified approach runs, labeled by outcome as binary hazard. All three anchor the label to the intervention or its outcome that occurs after the human has already recognized the hazard. So the label is late by at least the time the human takes to decide that intervention is necessary, and no intervention-supervised method annotates the earlier moment at which the human first judged continued autonomy unsafe.

C. Survival Analysis and Sequential Change Detection

Survival analysis models time to an event while retaining observations that end without the event as right-censored information, and in discrete time the hazard is the probability that the event occurs in an interval given that it has not occurred earlier [19], [20]. OSCAR [26] recently applied it to robot navigation with an obstacle clearing as the event, whereas CUSP takes the human-annotated perception onset as its event rather than the physical outcome that intervention makes unavailable.

CUSUM [21] detects sustained change by accumulating evidence for a post-change distribution and resetting when nominal evidence dominates, with classical optimality results under specified change models [27], [28]. CUSP neither treats its learned score as a likelihood ratio nor invokes these guarantees, but standardizes the score against the site's normal driving, accumulates sustained elevations, and selects the threshold empirically from that driving.

III. METHOD

CUSP attaches to an existing navigation model and requires no change to the policy or its training. A hazard head on frozen vision-foundation features is trained as a discrete-time survival model on the onset (Secs. III-C–III-E), and a one-sided CUSUM accumulates its runtime score under a site-local false-alarm budget (Sec. III-F).

A. Problem Definition

Notation. An *intervention* is the instant the human takes manual control, and only interventions driven by a hazard to the platform are *events*. Since intervention preempts the failure, every event is right-censored at intervention, and CUSP predicts the onset of human-perceived danger rather

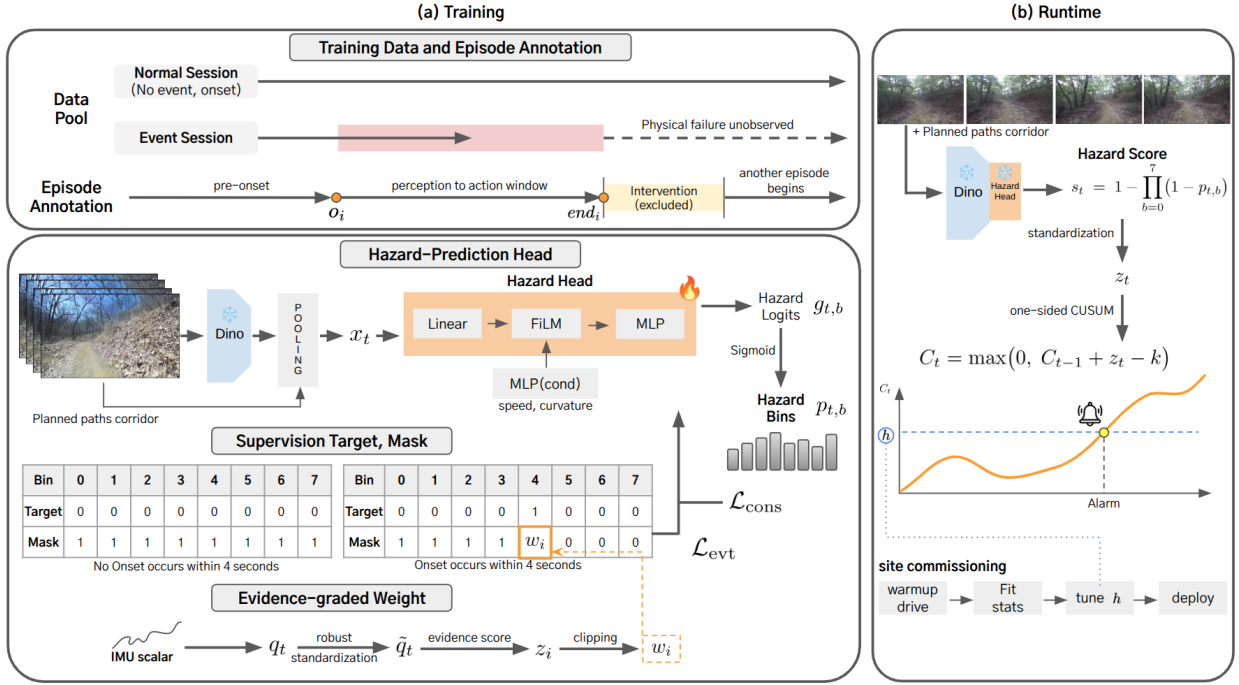


Fig. 2: CUSP overview. (a) Training. Each episode is annotated with the perception onset o_i and the end_i just before intervention. Frozen DINOv3 features are pooled over the committed-path corridor with temporal differences and global context and FiLM-conditioned on path curvature and speed. Survival targets and masks are anchored on o_i , and per-event weights w_i are graded by IMU tilt around the onset (training only). (b) Runtime. The onset-risk score is robust-standardized and accumulated by a one-sided CUSUM under a per-site false-alarm budget.

than the failure. CUSP samples the run at rate f , with frame t timestamped t/f . Frame t carries the camera image I_t , the committed path $\mathcal{P}_t$, its curvature $curv_t$, and the platform speed spd_t measured from pose, which are the only runtime inputs. Event i has an *onset* o_i , the earliest time at which continuing along $\mathcal{P}_t$ would lead into the hazard, and an *end* just before intervention, and $[o_i, end]$ is the *perception-to-action window*. The *horizon* H is how far ahead the alarm looks, discretized into $B=H/\Delta$ bins with $\Delta=1/f$. The interval $[o_i - H, o_i)$ is the *positive band*, *normal driving* comprises every frame outside the perception-to-action windows and the human-controlled segments, and a *site* is one physical environment.

Task. An *alarm* raised inside $[o_i, end]$ is a *detection* of event i , and an alarm during normal driving is a *false alarm*, counted per accumulation episode and reported as a rate per unit of normal driving. The *false-alarm budget* α is the maximum such rate a site tolerates, and it is the sole quantity the decision threshold is tuned to meet. CUSP must detect as many onsets as possible while keeping each site’s false-alarm rate within α , with one learned model that is not retrained per site.

B. Onset Annotation

Both timestamps are annotated post hoc from the recorded video. Annotators play each session in order, mark the frame just before each intervention as the end, then rewind and re-watch the approach to locate the first frame at which

continuing the committed path would lead into the hazard and mark its timestamp. Three annotators marked each onset, and the median of their timestamps defines the final label o_i . This aggregation limits the influence of a single early or late timestamp. The onset is therefore a retrospective judgment made knowing that an intervention followed. A parallel IMU stream grades each event’s label weight (Sec. III-D), and it is not used at runtime.

C. Input Features and Architecture

Visual input. The frozen DINOv3 [29] backbone encodes I_t into patch features of dimension $D=768$, pooled in two ways. f_t^{plan} is the mean over the patches that $\mathcal{P}_t$ covers when projected into I_t as a corridor of the platform’s width, and f_t^{glob} is the mean over all patches, so the former describes the terrain the plan is about to enter and the latter the scene as a whole. Both use only frame t . As the platform advances from frame t to $t + \ell$, the corridor feature changes by $\delta_{\ell} f_{t+\ell}^{\text{plan}} = f_{t+\ell}^{\text{plan}} - f_t^{\text{plan}}$, which is appended to the input of the later frame, so the alarm is causal. Writing t for the current frame, the visual input is

$$x_t = \text{concat}(f_t^{\text{plan}}, \delta_{\ell_1} f_t^{\text{plan}}, \delta_{\ell_2} f_t^{\text{plan}}, f_t^{\text{glob}}) \in \mathbb{R}^{4D}, \quad (1)$$

where $\ell_1 < \ell_2$ span about 1 and 2 s, so the head sees what lay ahead one and two seconds ago against what lies ahead now.

FiLM conditioning. The same terrain carries different risk when driven slowly and straight versus fast through a turn, so

curv_t and spd_t modulate the visual representation via FiLM,

$$\begin{aligned} h_0 &= \text{LN}(Wx_t), \\ (\gamma, \beta) &= \text{MLP}_{\text{cond}}([\text{curv}_t, \text{spd}_t]), \\ h_1 &= \phi(h_0 \odot (1 + \gamma) + \beta), \\ g_t &= W_3 \phi(W_2 h_1), \end{aligned} \quad (2)$$

applied once after the $4D \rightarrow 256$ projection and layer normalization, where ϕ is the SiLU activation and W, W_2, W_3 are linear layers. A $256 \rightarrow 128 \rightarrow B$ head produces the hazard logits $g_t = (g_{t,0}, \dots, g_{t,B-1})$ with $p_{t,b} = \sigma(g_{t,b})$.

D. Onset Target and Evidence-Graded Supervision

Hazard target. We set $H=4\text{ s}$ so that the horizon exceeds the 2.5 s perception–reaction time adopted as a design value for a driver facing an unexpected hazard [30], [31], leaving a margin between the predicted risk and the human’s own response. The head predicts for each bin b the conditional hazard $p_{t,b}$ that onset occurs in bin b given that it has not occurred earlier. With $d = o_i - t/f$, a frame in the positive band ($0 < d \leq H$) has its onset at bin

$$b_o = \min(\lfloor d/\Delta \rfloor, B-1) \quad (3)$$

and per-bin targets and masks

$$Y_{t,b} = \mathbf{1}[b = b_o] \quad M_{t,b} = \begin{cases} 1, & b < b_o \\ w_i, & b = b_o \\ 0, & b > b_o. \end{cases} \quad (4)$$

Pre-onset bins are supervised as “not yet,” the onset bin is the sole positive with weight w_i , and post-onset bins are masked because the event has already occurred. A frame with no onset within the horizon has $Y_{t,b} = 0$ and $M_{t,b} = 1$ for every b and provides only survival supervision. Frames in $[o_i, \text{end}]$ and the human-controlled segment after intervention are not used for training.

Evidence-graded weight w_i . The onset is annotated from video after the fact, so its timing varies with the annotator. Following prior work that weights human-provided supervision by a confidence score rather than using it uniformly [32], and in learning from interventions grades it by an independent physical criterion [33], [34], we weight each onset by IMU tilt inside the positive band. Hazards from robot–terrain interaction act on the platform by the time the human perceives them, so tilt shows, independently of the annotator, that the annotated moment lies on a physical process already under way, whereas an onset without tilt rests on the video alone. From IMU roll ϕ_t and pitch θ_t we form $q_t = \sqrt{\phi_t^2 + \theta_t^2}$ and robust-standardize it against normal driving,

$$\tilde{q}_t = \frac{q_t - \text{med}}{c \cdot \text{MAD} + \epsilon}, \quad (5)$$

with med and MAD over normal-driving frames, $c \approx 1.4826$ the consistency constant, and a small ϵ . The evidence score is the peak over the positive band,

$$z_i = \max_{o_i - H \leq t/f < o_i} \tilde{q}_t, \quad (6)$$

mapped to a clipped linear ramp between bounds $z_{lo} < z_{hi}$,

$$w_i = \text{clip}\left(\frac{z_i - z_{lo}}{z_{hi} - z_{lo}}, 0, 1\right). \quad (7)$$

With $z_{lo}=1$ and $z_{hi}=2$ an onset receives positive weight only once its peak tilt exceeds the central spread of normal driving and full weight at two robust standard deviations. An event with $w_i=0$ is neither discarded nor relabeled as normal driving. Its onset bin is excluded from the loss while its pre-onset bins retain $M_{t,b}=1$ and $Y_{t,b}=0$, so the event is right-censored at its onset and only the transition time is withheld. The same censoring covers hazards that induce no tilt before onset, such as a level approach to an obstacle, which the head still learns to score since the visual change of a corridor becoming undrivable is shared with the tilt-verified onsets on which it is trained.

E. Training Objective

Event loss. Positives are rare per bin, so with pos_b and neg_b the per-bin counts over the training set, the positive term of the binary cross-entropy is scaled by

$$\pi_b = \text{clip}\left(\frac{\text{neg}_b}{\max(\text{pos}_b, 1)}, 1, \pi_{\max}\right), \quad (8)$$

with $\pi_{\max}$ fixed in advance, and the event loss is the mask-normalized sum

$$\mathcal{L}_{\text{evt}} = \frac{\sum_{t,b} M_{t,b} \text{BCE}_{\pi_b}(g_{t,b}, Y_{t,b})}{\max(\sum_{t,b} M_{t,b}, 1)}. \quad (9)$$

Under Eq. (4) this is a weighted discrete-time survival negative log-likelihood, in which pre-onset bins supervise survival, the onset bin supervises the hazard, and post-onset bins leave the risk set through the mask.

Photometric consistency. We form x'_t from $\text{aug}(I_t)$ as in Eq. (1), with unchanged conditioning $c_t = [\text{curv}_t, \text{spd}_t]$, and

$$\mathcal{L}_{\text{cons}} = \text{mean} \|\text{head}(x_t, c_t) - \text{head}(x'_t, c_t)\|_2^2 \quad (10)$$

pushes the head toward appearance invariance and stabilizes the score scale across sites. The total objective is $\mathcal{L} = \mathcal{L}_{\text{evt}} + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}}$.

F. Runtime Score and CUSUM Decision

Runtime score. The decision layer needs one scalar from the B bin-wise hazards. Each bin survives with probability $1 - p_{t,b}$, so the complement of the product is the probability of at least one onset within the horizon,

$$s_t = 1 - \prod_{b=0}^{B-1} (1 - p_{t,b}). \quad (11)$$

Because of π_b the head is class-balanced rather than calibrated, so s_t is a ranking score and is standardized before any decision.

CUSUM. A hazard builds up over the approach rather than announcing itself in a single frame. The score is standardized against the site’s normal driving as in Eq. (5),

$$z_t = \frac{s_t - \text{med}_0}{c \cdot \text{MAD}_0 + \epsilon}, \quad (12)$$

and accumulated by a one-sided CUSUM [21],

$$C_t = \max(0, C_{t-1} + z_t - k), \quad C_0 = 0. \quad (13)$$

An alarm is raised at the first crossing $C_t \geq h$, after which the statistic resets and re-arms, and one such episode is one alarm. A frame adds to C_t only when its score lies more than k robust standard deviations above the normal-driving median, so isolated excursions decay and only sustained excess accumulates.

Threshold and per-site commissioning. We fix k across sites and set h per site from a short unlabeled warm-up drive. The warm-up gives $(\text{med}_0, \text{MAD}_0)$ and, through Eq. (13), a trace of C_t on normal driving. We take h as the smallest value that this trace crosses no more than α times per hour. Commissioning sets only $(\text{med}_0, \text{MAD}_0, h)$; the head and other parameters stay fixed.

IV. EXPERIMENTS

A. Experimental Setup

Platform. All data were collected on a Clearpath Husky A200 UGV with a forward-facing ZED 2i stereo camera and built-in IMU. The left RGB stream is the only runtime image input, and the IMU provides the training-time evidence. The alarm runs at $f=2$ Hz, and inference takes 10 ms per frame on a Jetson AGX Orin.

Data. All autonomous driving is by NoMaD [35], whose predicted trajectory serves as the committed path, with CUSP attached as an alarm that leaves the policy’s commands unchanged. Training uses 163 intervention-terminated events and 329 min of normal driving from six training sites. Testing uses five held-out sites that are physically separate from the training sites and from one another. The 163 is the number of onsets rather than of supervision signals. Under the survival objective all 39,480 image frames supervise the head, frames in a positive band with the interval in which the onset falls and all other frames as right-censored. Sites I–II (11/50 events, 30.4/42.7 min of normal driving) are driven by NoMaD, so their committed paths are on-policy, and Sites III–V (28/13/40 events, 9.3/49.0/73.6 min) are teleoperated.

Implementation details. The backbone is a frozen DINOv3 ViT-B/16 [29] on 224×224 frames, and the plan corridor samples the projected path at several lateral offsets spanning the platform width. The hazard head is a three-layer MLP ($4D \rightarrow 256 \rightarrow 128 \rightarrow B$, $B=8$) with LayerNorm, SiLU, and dropout 0.1, and a two-layer MLP produces the FiLM parameters from the z-scored pair $[\text{curv}_t, \text{spd}_t]$. Photometric augmentations are brightness, contrast, saturation, hue, gamma, and blur. Training uses AdamW (learning rate 8×10^{-4} , weight decay 2×10^{-3}), batch size 4096, and 70 epochs. The deployed alarm averages five heads from independent seeds, with $\ell_1=2$, $\ell_2=4$, $\lambda_{\text{cons}}=1$, $\pi_{\text{max}}=40$, $\epsilon=10^{-9}$, and $k=3$ for CUSP. All hyperparameters were fixed by leave-one-site-out cross-validation on the training sites, and no test-site data informed any design choice.

Kinematic conditioning. NoMaD provides the committed path at the training sites and Sites I–II. The teleoperated Sites III–V have no such path, so we reconstruct one from the teleoperator’s realized future trajectory in the vehicle

TABLE I: Baselines: native output $\rightarrow$ per-frame hazard score.

Method	Native output	Hazard score
<i>Traversability</i>		
STERLING [2]	terrain feature	roughness head
CAHSOR [8]	terrain feature	roughness head (speed-cond.)
SALON [36]	visual feature	Mahalanobis to normal
WVN [1]	roughness, confidence	roughness + RND novelty
<i>Obstacle proximity</i>		
NavDP [10]	trajectory critic	negated critic of worst sample
CARE [11]	repulsive force	peak force along plan
<i>Plan/policy-conditioned</i>		
LaND [17]	intervention prob.	logit
PAAD [18]	failure prob. along plan	logit
FIPER [14]	novelty, action spread	conformal p-value

frame using Hesai LiDAR-inertial odometry at 10 Hz. This gives curv_t as heading change per arc length, spd_t from pose differences clipped to $[0, 3]$ m/s, and the corridor for f_t^{plan} . Conditioning at Sites III–V is thus off-policy and hindsight-based, whereas Sites I–II use a genuine on-policy plan.

Metrics and commissioning. Event i is detected when an alarm is raised inside $[o_i, \text{end}]$, at time τ_i . We report:

- *Detection*: the number of events with such an alarm, overall and per site.
- *Margin*: $\text{end} - \tau_i$, reported as counts exceeding 0.5/1/2 s.
- *Delay*: $\tau_i - o_i$, reported as median with IQR over detected events.
- $C(+3)$: the fraction of all events with delay ≤ 3 s.

All methods use the same false-alarm budget $\alpha=10^{-1}$, with thresholds set from each site’s first 2 min of normal driving. Realized rates are measured over each site’s full normal driving.

Baselines. We compare using nine methods from three families (Table I). Each method’s final output is mapped to a per-frame hazard score with the method itself left unchanged. Every learned baseline is trained or fit on the training sites and, like CUSP, averages five heads from independent seeds; NavDP and CARE use their released weights, and STERLING and CAHSOR, which output a representation rather than a score, receive a two-layer head fit to roughness. LaND, PAAD, and FIPER run on-policy at Sites I–II; at Sites III–V, LaND and PAAD take the same hindsight conditioning as CUSP, and FIPER runs the policy open-loop in goal-masked exploration mode, as no goal is recorded. LaND and PAAD output probabilities near zero on most normal-driving frames (93% and 72%), which cannot form the tail an accumulating rule requires, so they are logit-transformed before standardization and each baseline is reported under the better of its raw and transformed score.

Downstream of the hazard score, every method uses robust standardization against the site’s normal driving and a threshold commissioned per method under the same target false-alarm budget. We compare CUSUM with $k \in \{1, 2, 3\}$ and the instantaneous rule per baseline; $k=3$ is already the best setting for six of the nine, including four of the five best-performing, and no baseline exceeds 26 detections under any setting, so every baseline is reported at its best configuration.

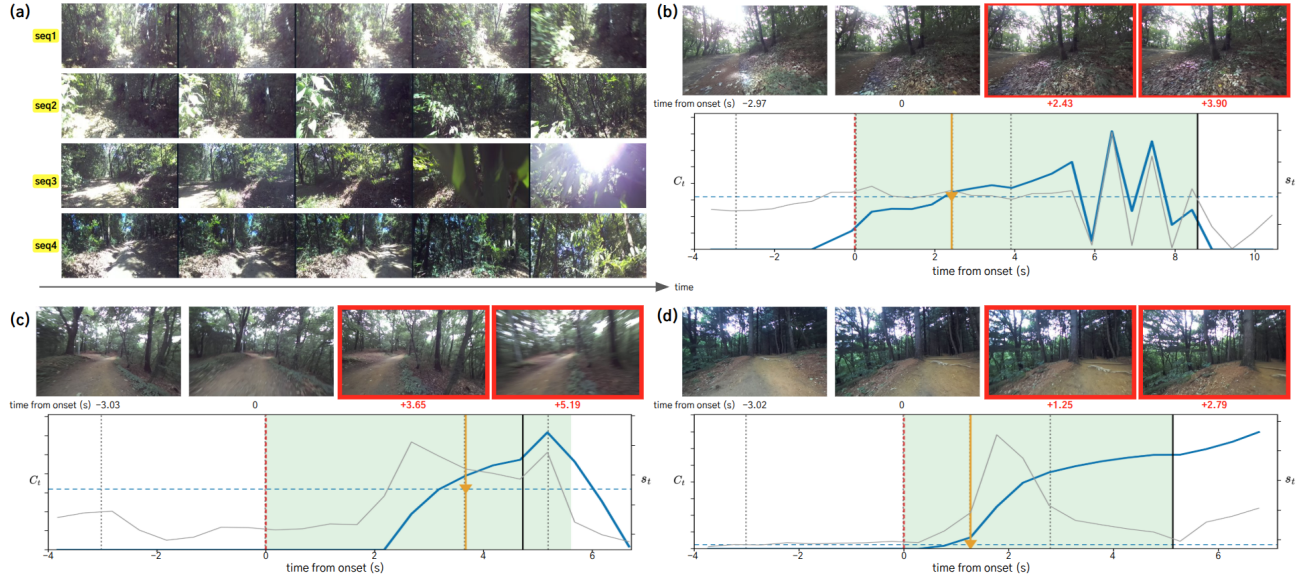


Fig. 3: Example detections at Sites II, III, and V. (a) Four vegetation intrusion events at Site II, where the path is visibly impassable and the baselines also detect the events. (b) Mounting a right cross-slope (Site V). (c) Sliding into a downward right cross-slope (Site III). (d) Entering a left cross-slope overgrown with dense bushes (Site III). In (b)–(d) nothing in the view marks the path as impassable, and CUSP is the only method that alarms. Each plot shows the score s_t (grey) and the CUSUM statistic C_t (blue) against time from onset, with the threshold h as a dashed line, the perception-to-action window $[o_i, \text{end}]$ shaded in green, and the alarm time marked by the orange vertical line. Frames above each plot are sampled at the indicated times from onset.

TABLE II: Five-site results under a common 10/h budget.

Method events	det. ($\uparrow$)/142 (I–V) 142 (11/50/28/13/40)	margin ($\uparrow$)>0.5/1/2 (s)	$C_t(+3)$ ($\uparrow$)	delay ($\downarrow$) [IQR] (s)
NavDP [10]	3 (0/3/0/0/0)	3/2/2	0.014	+2.04 [1.8–5.7]
CARE [11]	10 (0/5/1/1/3)	10/10/10	0.021	+4.59 [1.7–5.7]
LaND [17]	12 (0/0/3/3/6)	11/11/10	0.049	+2.58 [1.4–3.8]
FIPER [14]	21 (1/10/6/0/4)	19/16/12	0.028	+5.24 [3.2–8.4]
PAAD [18]	26 (0/14/3/1/8)	25/22/19	0.056	+5.13 [2.8–7.7]
STERLING [2]	7 (1/5/0/1/0)	7/6/5	0.000	+5.21 [4.4–6.0]
CAHSOR [8]	9 (2/6/1/0/0)	8/7/6	0.014	+5.60 [3.4–9.4]
SALON [36]	24 (0/20/2/2/0)	19/17/13	0.028	+4.69 [3.2–8.4]
WVN [1]	26 (1/16/2/3/4)	21/20/17	0.063	+5.00 [2.5–7.4]
CUSP (intervention label)	34 (3/17/8/4/2)	34/32/24	0.127	+2.91 [1.6–5.4]
CUSP (ours)	85 (4/40/16/4/21)	77/72/64	0.218	+3.72 [1.9–6.5]

B. Results

Table II reports all five sites under a common 10/h budget, realized at 7.9/h at Site I, 9.8/h at Site II, and 9.5/h on Sites III–V pooled (Table VI). CUSP detects 85 of 142 events, 3.3 times the 26 of the two strongest baselines, and leads at every site. The lead holds under both autonomous and teleoperated driving: on the on-policy Sites I–II, where CUSP ran on top of NoMaD while leaving its commands unchanged, it detects 44 of 61 events against 20 for the best baseline, and on the teleoperated Sites III–V 41 of 81 against 12. It also detects the largest fraction within 3 s of onset: 22% of all events against at most 6.3% for any baseline, and 64 of the 85 detections leave more than 2 s before intervention.

The detections split by hazard type. Where the hazard is a visible obstruction, as at Site II where the policy drives into bushes and tree trunks that fill the corridor, the baselines also raise alarms and concentrate their detections there. However, CUSP still detects 40 of 50 events against 20 for the best of them (Fig. 3(a)). WVN, which combines roughness with novelty, detects 26 events overall compared with 7 for STERLING. At the other sites the path ahead looks drivable, baseline detections are fewer, and the danger comes from cross-slopes and banks acting on the platform. The question is when continuing becomes unsafe rather than whether the path is passable. The margin is widest (Fig. 3(b)–(d)), and the lead holds at every budget on Sites III–V (Fig. 4): the detection rate the best baseline reaches at the 10/h budget, CUSP reaches at under 3/h, and the 0.30 the best baseline reaches at 30/h, CUSP reaches at 6/h.

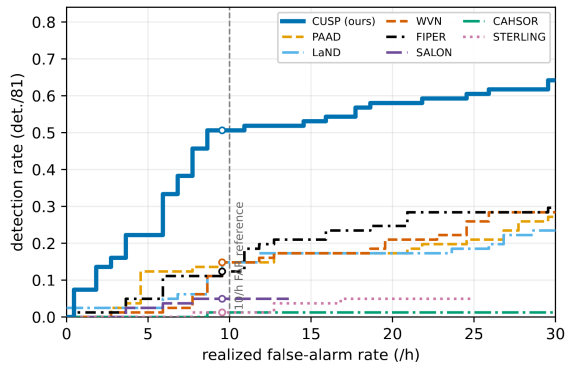


Fig. 4: Detection rate against realized FAR on Sites III–V as h varies. The dashed line marks the 10/h operating point.

TABLE III: Training objective ablation.

Objective	Hor.	Mask	w_i	det./142 (I-V)	Δ [95% CI]
Binary [17], [18]	H	—	—	9 (2/1/3/0/3)	-76 [-89, -64]
Per-bin binary	B bins	no	—	4 (0/1/1/0/2)	-81 [-92, -70]
Time-to-onset regr.	cont.	—	—	11 (0/3/2/1/5)	-74 [-85, -63]
Survival	B bins	yes	uniform	45 (0/8/15/3/19)	-40 [-49, -32]
Survival (CUSP)	B bins	yes	graded	85 (4/40/16/4/21)	—

TABLE IV: Positive label extended to $[o_i, o_i + \eta]$.

η (s)	0	+1	+2	+4	+6	+10	end
det./142	85	72	72	69	62	60	58
I/II	4/40	0/31	1/33	3/30	2/30	2/29	2/28
III/IV/V	16/4/21	16/5/20	15/4/19	13/4/19	9/4/17	10/4/15	8/5/15

C. Ablation Studies

For detection-count differences (variant minus CUSP), we report 95% percentile CIs from 4,000 site-stratified paired bootstrap replicates, resampling events with replacement within each site while preserving paired predictions. Each replicate retains the observed event count per site and recomputes the pooled detection-count difference. Bounds are the 2.5th and 97.5th percentiles.

Intervention label (Table II). Training the same head on intervention instead of onset lowers detection, reflecting the different positive classes. Anchored at the intervention, the positive band $[\text{end}-H, \text{end})$ consists of frames already inside the hazard and separates from normal driving far more cleanly than the onset band $[o_i-H, o_i)$. So, the head learns to output scores near one, whereas the onset-trained head remains moderate even inside events. Neither head avoids occasional false outputs on normal driving, and the two differ not in how often such outputs occur but in how large they are. The intervention-trained head produces them at the magnitude it learned for true events, whereas the onset-trained head, never trained to be confident, produces small ones. Standardization turns each large false output into a very large z_t , which the CUSUM holds above h for many frames, so a handful of false outputs uses up much of the false-alarm budget and fewer events are detected at the common budget. The onset label trains a harder class, and in return it bounds false outputs so that the alarm releases between events.

Training objective (Table III). With features, head, consistency term, and decision layer fixed, we isolate the choice of supervision. Binary classification [17], [18] detects 9 events, while per-bin classification and time-to-onset regression detect 4 and 11. The survival mask (Eq. (4)) raises detection to 45, and evidence grading to 85. These comparisons support modeling onset as a time-to-event target rather than classifying the same visual inputs.

Positive-bin extent (Table IV). Supervising the transition is more effective than supervising an ongoing hazard. Extending the positive label to $[o_i, o_i + \eta]$, capped at event end, lowers detection at every offset, from 85 to 72 at $\eta=1$ s and to 58 at event end, with the largest losses at Sites II and III, where the perception-to-action windows are longest.

TABLE V: Evidence-weight mapping $z_i \rightarrow w_i$.

Scheme	w_i	$w_i > 0/163$	det./142 (I-V)	Δ [95% CI]
Soft ramp	$\text{clip}(z_i/2, 0, 1)$	146	41 (0/3/15/5/18)	-44 [-51, -36]
Uniform (evidence off)	1	163	45 (0/8/15/3/19)	-40 [-49, -32]
Sigmoid	$\sigma(z_i - 1)$	163	48 (0/9/15/4/20)	-37 [-46, -28]
Gate	$\mathbf{1}[z_i \geq 1.5]$	89	62 (2/23/14/5/18)	-23 [-32, -14]
Clipped ramp (CUSP)	Eq. (7), (1, 2)	114	85 (4/40/16/4/21)	—

TABLE VI: Decision statistic on the frozen CUSP score. Site- σ is the population SD of the five FARs.

Decision statistic	det./142 (I-V)	Δ [95% CI]	site- σ (/h)	FAR (/h, I-V)
Conformal martingale	16 (1/3/4/5/3)	-69 [-81, -57]	5.9	5.9/8.4/6.4/19.6/2.4
Sliding sum	26 (2/2/8/7/7)	-59 [-69, -48]	3.3	7.9/9.8/12.9/14.7/5.7
EWMA	37 (2/12/8/6/9)	-48 [-59, -38]	4.0	13.8/9.8/6.4/15.9/5.7
Instantaneous	53 (2/20/10/6/15)	-32 [-43, -22]	6.4	17.8/9.8/0.0/15.9/6.5
CUSUM (CUSP)	85 (4/40/16/4/21)	—	2.2	7.9/9.8/12.9/12.2/7.3

TABLE VII: Kinematic conditioning on autonomous Sites I–II.

Conditioning	det./61 (I/II)	Δ [95% CI]	FAR (/h) [95% upper]
FiLM off	8 (0/8)	-36 [-43, -28]	8.2 [≤ 18.2]
concat. [curv, spd]	36 (0/36)	-8 [-14, -3]	8.2 [≤ 19.5]
FiLM (CUSP)	44 (4/40)	—	9.0 [≤ 19.5]

Evidence-weight shape (Table V). The decisive choice is censoring, giving zero weight to onsets without tilt evidence. Keeping all or nearly all onsets detects 41–48 events; gating raises this to 62, and grading with the clipped ramp to 85. Censoring does not raise the score on average but sharpens its upper tail, and only frames more than k robust standard deviations above the normal-driving median increase C_t . At Site II the events accumulated to about half of h under uniform weighting and to about twice h under the clipped ramp, so the sharper tail moved most of them across the threshold. The largest sensitivity to the mapping occurs at Site II, where hazards are visible obstructions on level ground, so censoring tilt-less onsets does not sacrifice that hazard type.

Decision statistic (Table VI). All five rules run on the same z_t under the same commissioning. The instantaneous statistic thresholds z_t directly, the sliding sum adds z_t over $W=B$ frames as in FIPER [14], the EWMA uses $\lambda=0.3$, and the conformal martingale [37] thresholds the power martingale of z_t p-values with $\varepsilon=0.5$. CUSUM detects the most events because it treats normal driving and hazards differently. On normal driving about 97% of frames lie below the drift k and do not increase C_t ; negative increments reduce accumulated evidence toward zero. On a hazard the score rises weakly but persistently over several seconds, and accumulation turns that persistence into amplitude, so 60% of events reach h against 37% for the memoryless instantaneous statistic. CUSUM also has the lowest between-site dispersion of realized FARs, whereas the instantaneous and EWMA statistics exceed the budget at Sites I and IV and fall below budget at Sites III and V.

Kinematic conditioning (Table VII). On the autonomous Sites I–II, with each variant re-commissioned under the same budget, removing FiLM detects 8 events and plain

concatenation of $[\text{curv}_t, \text{spd}_t]$ detects 36, against 44 for FiLM at a comparable realized rate. Neither alternative detects any events at Site I. Speed and curvature therefore help most when they modulate the visual representation.

V. CONCLUSION AND FUTURE WORK

We introduced CUSP, a runtime hazard alarm for off-road navigation. We treat the human-perceived onset, annotated separately from intervention, as a time-to-event target. A discrete-time survival model learns from observed onsets and driving without onsets in intervention-terminated logs, and CUSUM converts onset risk into alarms calibrated from normal driving. CUSP requires no failure data or policy changes. Across five unseen sites under a common false-alarm budget, it detected 85 of 142 events against 26 for the best of nine adapted baselines, with the largest gains on hazards that arise from driving the terrain rather than from a visible obstruction.

Limitations and future work. Three limitations point to the next steps. First, what CUSP can detect is bounded by the hazard types the training onsets span. Coverage saturates at about 60% of the events on the teleoperated Sites III–V, where raising the budget only adds false alarms (Fig. 4), and at about 80% on Site II, so the remaining misses may reflect hazards underrepresented in training. Second, the alarm sees only what the low, narrow platform-mounted camera sees, which leaves a ditch behind a crest or a berm below the near horizon without image evidence until the platform is on it, so a wider or elevated field of view is the next step. Third, CUSP is only a detector and leaves recovery, slowing, or re-planning to the human or the navigation model; the human’s actions inside the recorded perception-to-action window are a natural supervision source for that response, and learning it from the same logs is the next step.

REFERENCES

- [1] J. Frey, M. Mattamala, N. Chebrolu, C. Cadena, M. Fallon, and M. Hutter, “Fast traversability estimation for wild visual navigation,” in *Proc. Robot.: Sci. Syst. (RSS)*, 2023.
- [2] H. Karnan *et al.*, “STERLING: Self-supervised terrain representation learning from unconstrained robot experience,” in *Proc. Conf. Robot Learn. (CoRL)*, 2023.
- [3] G. Kahn, P. Abbeel, and S. Levine, “BADGR: An autonomous self-supervised learning-based navigation system,” *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 1312–1319, 2021.
- [4] M. V. Gasparino *et al.*, “WayFAST: Navigation with predictive traversability in the field,” *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 10651–10658, 2022.
- [5] A. J. Sathiamoorthy *et al.*, “TerraPN: Unstructured terrain navigation using online self-supervised learning,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2022, pp. 7197–7204.
- [6] X. Cai *et al.*, “EVORA: Deep evidential traversability learning for risk-aware off-road autonomy,” *IEEE Trans. Robot.*, vol. 40, pp. 3756–3777, 2024.
- [7] X. Cai *et al.*, “PIETRA: Physics-informed evidential learning for traversing out-of-distribution terrain,” *IEEE Robot. Autom. Lett.*, vol. 10, no. 3, pp. 2359–2366, 2025.
- [8] A. Pokhrel, A. Datar, M. Nazeri, and X. Xiao, “CAHSOR: Competence-aware high-speed off-road ground navigation in SE(3),” *IEEE Robot. Autom. Lett.*, vol. 9, no. 11, 2024.
- [9] D. D. Fan *et al.*, “STEP: Stochastic traversability evaluation and planning for risk-aware off-road navigation,” in *Proc. Robot.: Sci. Syst. (RSS)*, 2021.
- [10] W. Cai *et al.*, “NavDP: Learning sim-to-real navigation diffusion policy with privileged information guidance,” arXiv:2505.08712, 2025.
- [11] J. Kim, J. Sim, W. Kim, K. P. Sycara, and C. Nam, “CARE: Enhancing safety of visual navigation through collision avoidance via repulsive estimation,” in *Proc. Conf. Robot Learn. (CoRL)*, 2025, pp. 3704–3719.
- [12] R. Sinha, A. Elhafsi, C. Agia, M. Foutter, E. Schmerling, and M. Pavone, “Real-time anomaly detection and reactive planning with large language models,” in *Proc. Robot.: Sci. Syst. (RSS)*, 2024.
- [13] C. Agia *et al.*, “Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress,” in *Proc. Conf. Robot Learn. (CoRL)*, 2024.
- [14] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, “Failure prediction at runtime for generative robot policies,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2025.
- [15] Q. Gu *et al.*, “SAFE: Multitask failure detection for vision–language–action models,” in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2025.
- [16] C. Xu *et al.*, “Can we detect failures without failure data? Uncertainty-aware runtime failure detection for imitation learning policies,” in *Proc. Robot.: Sci. Syst. (RSS)*, 2025.
- [17] G. Kahn, P. Abbeel, and S. Levine, “LaND: Learning to navigate from disengagements,” *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 1872–1879, 2021.
- [18] T. Ji, A. N. Sivakumar, G. Chowdhary, and K. Driggs-Campbell, “Proactive anomaly detection for robot navigation with multi-sensor fusion,” *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 4975–4982, 2022.
- [19] K. Suresh, C. Severn, and D. Ghosh, “Survival prediction models: An introduction to discrete-time modeling,” *BMC Med. Res. Methodol.*, vol. 22, art. 207, 2022.
- [20] Z. Yang, J. Kannianen, T. Krogerus, and F. Emmert-Streib, “Prognostic modeling of predictive maintenance with survival analysis for mobile work equipment,” *Sci. Rep.*, vol. 12, art. 8529, 2022.
- [21] E. S. Page, “Continuous inspection schemes,” *Biometrika*, vol. 41, pp. 100–115, 1954.
- [22] Y. Xu, C. Qiu, Y. Wang, W. Wu, and Z. Li, “GAFT: Geo-anchored fine-tuning for hazard identification from rare failures,” arXiv:2608.30858, 2026.
- [23] M. Vukobratović and B. Borovac, “Zero-moment point—thirty five years of its life,” *Int. J. Humanoid Robot.*, vol. 1, no. 1, pp. 157–173, 2004.
- [24] A. Naik *et al.*, “Safety monitor for off-road planning with uncertainty bounded Bekker costs,” SAE Technical Paper 2026-01-0141, 2026.
- [25] L. Wellhausen, R. Ranftl, and M. Hutter, “Safe robot navigation via multi-modal anomaly detection,” *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 1326–1333, 2020.
- [26] H. Sahak, A. Jiao, N. Rhinehart, and T. D. Barfoot, “OSCAR: Obstacle survival curves for adaptive robot navigation,” arXiv:2606.00990, 2026.
- [27] G. Lorden, “Procedures for reacting to a change in distribution,” *Ann. Math. Statist.*, vol. 42, no. 6, pp. 1897–1908, 1971.
- [28] G. V. Moustakides, “Optimal stopping times for detecting changes in distributions,” *Ann. Statist.*, vol. 14, no. 4, pp. 1379–1387, 1986.
- [29] O. Siméoni *et al.*, “DINOv3,” arXiv:2508.10104, 2025.
- [30] M. Green, “How long does it take to stop? Methodological analysis of driver perception-brake times,” *Transp. Hum. Factors*, vol. 2, no. 3, pp. 195–216, 2000.
- [31] AASHTO, *A Policy on Geometric Design of Highways and Streets*, 7th ed. Washington, DC, USA: Amer. Assoc. State Highway Transp. Officials, 2018.
- [32] Y.-H. Wu, N. Charoenphakdee, H. Bao, V. Tangkaratt, and M. Sugiyama, “Imitation learning from imperfect demonstration,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 6818–6827.
- [33] W. Huang, H. Liu, Z. Huang, and C. Lv, “Safety-aware human-in-the-loop reinforcement learning with shared control for autonomous driving,” *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 16181–16192, 2024.
- [34] W. Huang, Z. Shan, S. Lou, and C. Lv, “Uncertainty-aware reinforcement learning for autonomous driving with multimodal digital driver guidance,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.
- [35] A. Sridhar, D. Shah, C. Glossop, and S. Levine, “NoMaD: Goal masked diffusion policies for navigation and exploration,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.
- [36] M. Sivaprakasam *et al.*, “SALON: Self-supervised adaptive learning for off-road navigation,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2025.
- [37] V. Vovk, I. Nouretdinov, and A. Gammerman, “Testing exchangeability on-line,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2003, pp. 768–775.